

Required Capstone Project: Module 7

Due Jul 15 by 11:59pm**Points** None**Available** after Jul 8 at 12am

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🎯 Learning Outcomes Addressed

- Understand how AI, GenAI and Agentic AI work and where they apply in business.
- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 3 – Agentic AI Design & Control

Milestone 7: Human-in-the-Loop & Responsibility Matrix

Estimated Effort: 1 Hours

Objective

In the previous module, you designed an **Agent Architecture Blueprint** that enables AI agents to plan and execute workflow tasks.

However, most enterprise AI systems **cannot operate fully autonomously** due to risks related to:

- financial impact
- regulatory compliance
- customer experience
- operational accountability

As a result, organisations implement **Human-in-the-Loop (HITL)** systems.

In these systems:

- AI performs automated tasks
- humans maintain **oversight, approvals, and final accountability**

In this module, you will design a **Human-in-the-Loop governance model** for the AI-enabled workflow developed earlier in the capstone.

You will define:

- where AI operates independently
- where human approvals are required
- how exceptions are handled
- who is responsible for outcomes

The goal is to design a system that balances **automation efficiency with operational control**.

How This Connects to Earlier Modules

Your submission should build on work completed earlier in the capstone:

- **Module 2** – You documented the **baseline workflow and risks**
- **Module 5** – You redesigned the **AI-enabled future workflow**
- **Module 6** – You designed an **AI agent architecture**

In this module, you will determine **how humans supervise and control the AI system operating within that workflow**.

Note: This is a required assignment and counts towards programme completion.

Task Overview

Using the workflow and AI agent designed in earlier modules, create a **Human-in-the-Loop Responsibility Matrix** that explains how humans and AI collaborate.

Your governance design should define:

- which workflow steps are **automated by AI**
- which steps require **human review or approval**
- how **errors or exceptions are escalated**
- how humans can **override AI outputs when required**

This structure ensures the AI system operates **safely and responsibly within business operations**.

Step 1 — Identify Automated vs Human-Controlled Steps

Start by reviewing your **AI-enabled workflow from Module 5**.

Determine which steps are handled by:

- AI automation
- human operators
- collaborative human–AI interaction

Some steps may remain fully human-controlled due to **risk, complexity, or regulatory requirements**.

Example (Illustrative)

Workflow: Customer support request processing

Workflow Step	Responsible Party
Receive customer request	Automated system
Analyse request content	AI
Draft response	AI
Review response	Human agent
Send response to customer	Human approval

This step clarifies **where automation occurs and where human oversight remains**.

Step 2 — Define Human Approval Checkpoints

Some decisions in the workflow may require **explicit human approval** before execution.

Approval checkpoints help reduce risk in areas such as:

- financial transactions
- regulatory compliance
- customer-facing communications
- high-impact operational decisions

Identify where these approvals should occur.

Example (Illustrative)

Approval Checkpoints

AI Output	Human Approval Required
Draft customer response	Yes
Knowledge article recommendation	Optional
Issue categorisation	No

This structure prevents **high-risk decisions** from being executed automatically.

Step 3 — Define Escalation Rules

Even well-designed AI systems may produce:

- uncertain outputs
- conflicting recommendations
- incomplete information

Escalation rules define **what happens when AI cannot complete a task reliably**.

Escalation may occur when:

- confidence scores fall below a defined threshold
 - the system detects ambiguous input
 - the task involves a sensitive decision
-

Example (Illustrative)

Escalation Rule Example

If AI confidence in issue categorisation falls below **80%**, the case is automatically routed to a **human support specialist** for manual review.

Escalation rules ensure that **uncertain cases are handled safely**.

Step 4 — Define Override Mechanisms

Human operators should always have the ability to **override AI decisions when necessary**.

Override mechanisms allow employees to:

- modify AI-generated outputs
- reject AI recommendations
- manually correct system actions

This ensures that **human judgement remains available when needed**.

Example (Illustrative)

Override Mechanism

Support agents can edit AI-generated response drafts before sending them to customers.

The system logs both the **original AI output and the final human-edited version** for auditing purposes.

Step 5 — Create a Responsibility Matrix

Finally, summarise the governance structure using a **Responsibility Matrix**.

This matrix clarifies **who is responsible for each decision point in the workflow**.

You may use a **RACI-style structure**.

Example (Illustrative)

Workflow Activity	AI System	Operations Team	Manager
Request categorisation	Responsible	Review if needed	—
Response generation	Responsible	Approve	—
Complex case handling	Assist	Responsible	—
Policy escalation	—	Responsible	Accountable

This matrix provides a clear overview of **decision ownership and accountability**.

Required Output Structure

Your **Human-in-the-Loop Responsibility Matrix** should include the following sections.

1. Automated vs Human-Controlled Workflow Steps

Identify which parts of the workflow are handled by AI and which remain human-controlled.

2. Human Approval Checkpoints

Define where human review or approval is required before actions are executed.

3. Escalation Rules

Explain how the system handles:

- uncertain outputs
- complex cases
- operational failures

4. Override Mechanisms

Describe how humans can modify or reject AI outputs.

5. Responsibility Matrix

Provide a simple table showing who is responsible for:

- AI system operations
- human review tasks
- final accountability

Practical Tips

Focus on Risk-Based Decisions

Human oversight is most important for **high-impact decisions**, such as:

- financial approvals
- compliance-sensitive actions
- customer-facing communications

Avoid Over-Escalation

If every AI output requires human review, the automation benefit disappears.

Design oversight **strategically rather than universally**.

Maintain Clear Accountability

Even when AI performs tasks, **humans remain responsible for outcomes**.

Define clear ownership for critical decisions.

Avoid These Common Mistakes

- ✗ Removing human oversight entirely
- ✗ Escalating every decision unnecessarily
- ✗ Designing AI workflows without accountability
- ✗ Failing to define who approves high-risk outputs

Effective AI systems require **balanced automation and controlled governance**.

Submission

Before submitting your **Human-in-the-Loop Responsibility Matrix**, confirm that your document includes:

- ✓ Automated vs human-controlled workflow steps
- ✓ Human approval checkpoints
- ✓ Escalation rules for uncertain outputs
- ✓ Override mechanisms for human intervention
- ✓ A responsibility matrix defining decision ownership

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 10

Criteria	Full Points	0 Points
Automated vs Human-Controlled Workflow Steps (2 points)	The submission identifies which workflow steps are handled by AI and which remain human-controlled. (2 points)	Workflow steps are not defined. (0 points)
Human Approval Checkpoints (2 points)	The submission defines where human review or approval is required before actions are executed. (2 points)	Human approval checkpoints are missing. (0 points)
Escalation Rules (2 points)	The submission explains how the system handles uncertain outputs, complex cases and operational failures with clear escalation logic. (2 points)	Escalation rules are missing. (0 points)
Override Mechanisms (2 points)	The submission describes how humans can modify, override or reject AI outputs. (2 points)	Override mechanisms are missing. (0 points)
Responsibility Matrix (2 points)	The submission includes a responsibility matrix showing ownership of AI system operations, human review tasks and final accountability. (2 points)	Responsibility matrix is missing. (0 points)

Why this Step Matters

As organisations adopt AI systems and agents, **governance becomes critical**.

Without clear oversight structures, AI deployments can introduce risks related to:

- incorrect decisions
- compliance failures
- operational accountability

Human-in-the-Loop systems ensure that AI remains:

- controlled
- auditable
- aligned with organisational responsibility structures

In the next module, you will explore how **Agentic AI can create disruptive business transformations**, moving beyond efficiency improvements towards **entirely new operating models**.